Top2018 MC Dataset — General Statistics and Geometric Analysis

Residues: 2,963,303 | **Chains:** 13,677 | **Structures:** 13,308 | **Source:** top2018mc_measures.jsonl | **Generated by:** pydangle-biopython v0.5.1

This document provides the following analyses:

1. **General Information** — dataset overview, measurements available, and summary statistics
2. **Amino Acid Composition and Chain Length Distribution** — comparison with UniProt and PDB reference databases
3. **Torsion Angle Distributions** — univariate (backbone, omega deviation, sidechain) and multivariate (Ramachandran) distributions using circular statistics
4. **Bond Angle Distributions** — tau (N−Cα−C) and other non-periodic angles using linear statistics
5. **References**

1. General Information

Dataset	Top2018 MC
Total residues	2,963,303
Unique structures	13,308
Unique chains	13,677

Measurements available: phi: 2,771,857, psi: 2,771,715, omega: 2,771,861, tau: 2,963,289, none. Not all residues have all measurements — terminal residues lack phi/psi/omega, and glycine and alanine lack sidechain chi angles.

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Ramachandran category distribution:
  General      1,841,337  (62.14%)
  IleVal        350,387  (11.82%)
  Gly           202,952  ( 6.85%)
  TransPro      112,742  ( 3.80%)
  PrePro         89,121  ( 3.01%)
  CisPro          5,949  ( 0.20%)

Torsion angles (circular statistics):
  Angle             N    Circ Mean     Circ Std
  -------- ---------- ------------ ------------
  phi       2,771,857      -85.137       41.703
  psi       2,771,715       11.214      114.636
  omega     2,771,861      179.205        8.002

Bond angles (linear statistics):
  tau       2,963,289      111.140        2.506
```

2. Amino Acid Composition and Chain Length Distribution

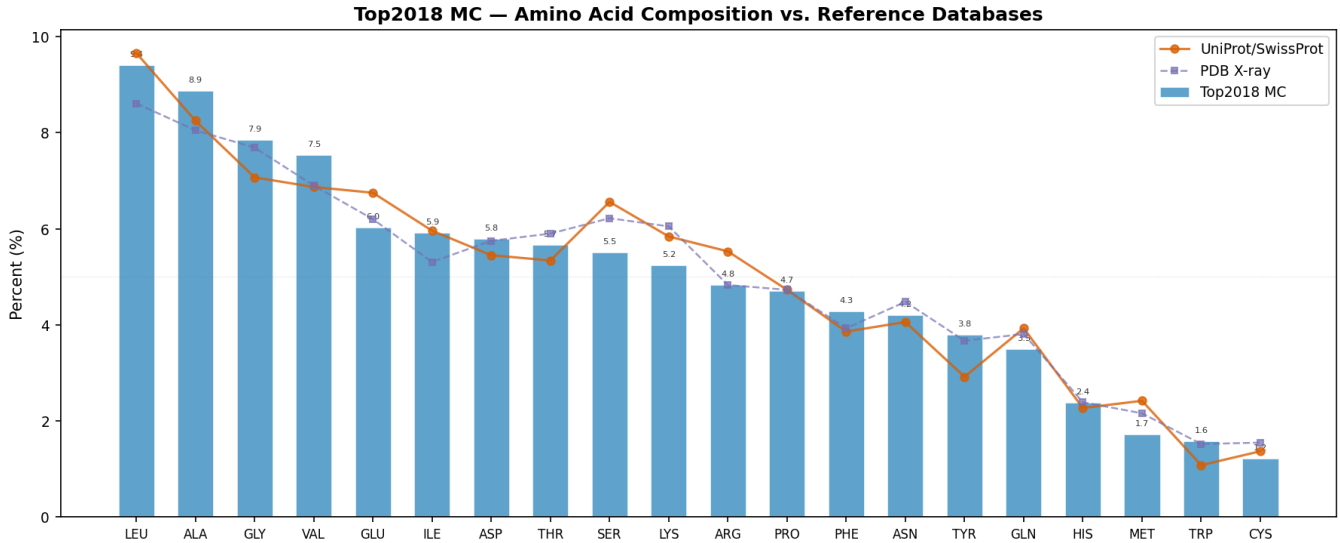
2.1 Representativeness

The figure and table below compare the Top2018 MC amino acid frequencies against two reference databases:

- **UniProt/SwissProt** (~2024): amino acid frequencies across all reviewed protein sequences, representing the broadest available view of protein sequence space.
- **PDB X-ray** (~2025): frequencies from a 10,000-entity sample of X-ray crystallographic structures in the RCSB Protein Data Bank.

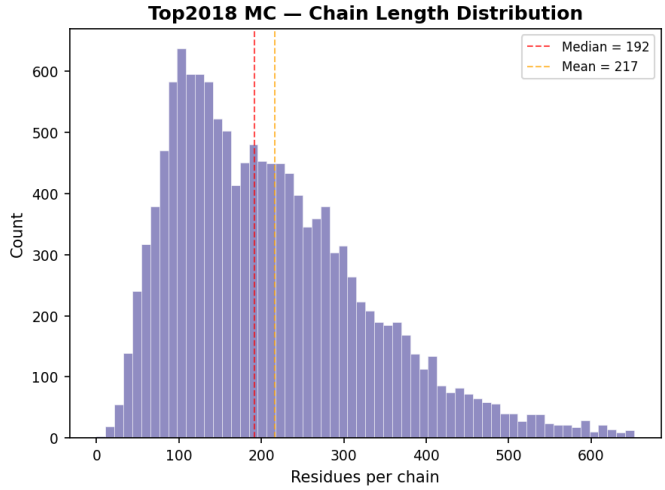
The two reference distributions are highly similar (most amino acids differ by <0.5%), indicating that the PDB's well-known crystallization bias — overrepresentation of soluble, globular, well-expressing proteins from model organisms — has only a modest effect on overall amino acid composition.

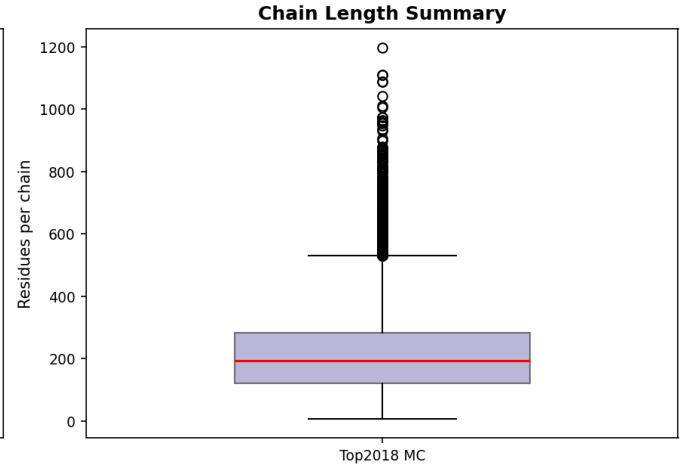
The Top2018 MC dataset (2,963,303 residues from 13,308 structures) shows additional deviations from both references due to quality filtering, homology reduction, and sample size effects.



Residue	Count	Dataset%	UniProt%	PDB%	Δ UniProt
LEU	278,872	9.41%	9.66%	8.61%	-0.25%
ALA	262,777	8.87%	8.25%	8.05%	+0.62%
GLY	232,691	7.85%	7.07%	7.69%	+0.78%
VAL	223,447	7.54%	6.87%	6.90%	+0.67%
GLU	178,620	6.03%	6.75%	6.20%	-0.72%
ILE	175,347	5.92%	5.96%	5.31%	-0.04%
ASP	171,682	5.79%	5.45%	5.75%	+0.34%
THR	167,699	5.66%	5.34%	5.90%	+0.32%
SER	163,093	5.50%	6.56%	6.22%	-1.06%
LYS	155,314	5.24%	5.84%	6.05%	-0.60%
ARG	143,215	4.83%	5.53%	4.83%	-0.70%
PRO	139,649	4.71%	4.73%	4.73%	-0.02%
PHE	126,640	4.27%	3.86%	3.93%	+0.41%
ASN	124,440	4.20%	4.06%	4.48%	+0.14%
TYR	112,200	3.79%	2.92%	3.67%	+0.87%
GLN	103,773	3.50%	3.93%	3.81%	-0.43%
HIS	70,358	2.37%	2.27%	2.39%	+0.10%
MET	50,790	1.71%	2.42%	2.16%	-0.71%
TRP	46,657	1.57%	1.08%	1.52%	+0.49%
CYS	36,039	1.22%	1.37%	1.55%	-0.15%

2.2 Chain Length Distribution



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Chain length statistics (13,677 chains):
  Min:              5
  Q1:             119
  Median:         192
  Mean:         216.7
  Q3:             283
  Max:          1,198
  Std:          130.3
```
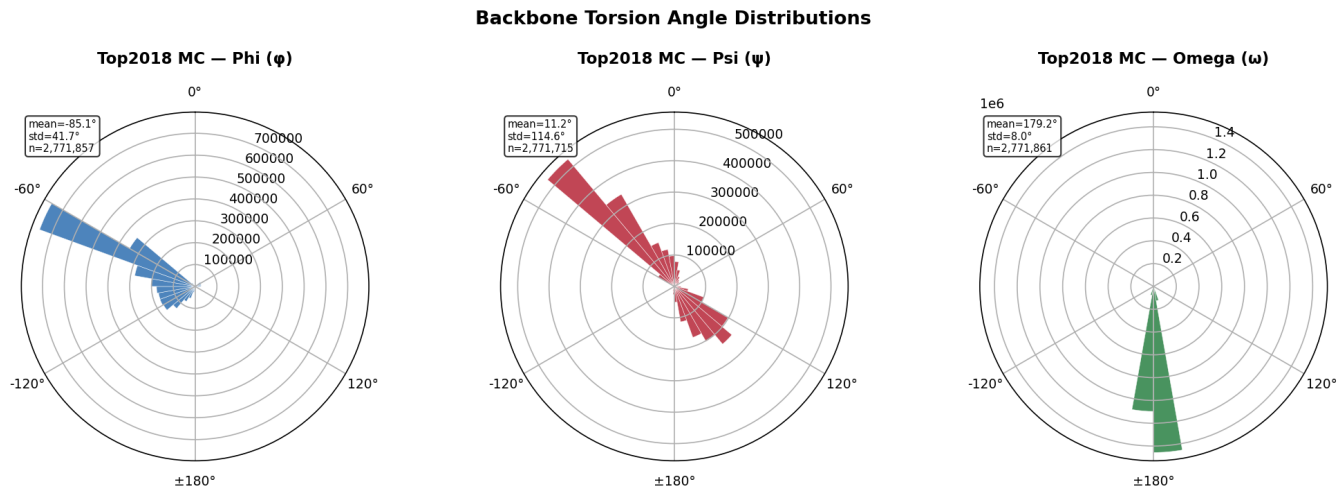

3. Torsion Angle Distributions (Circular Statistics)

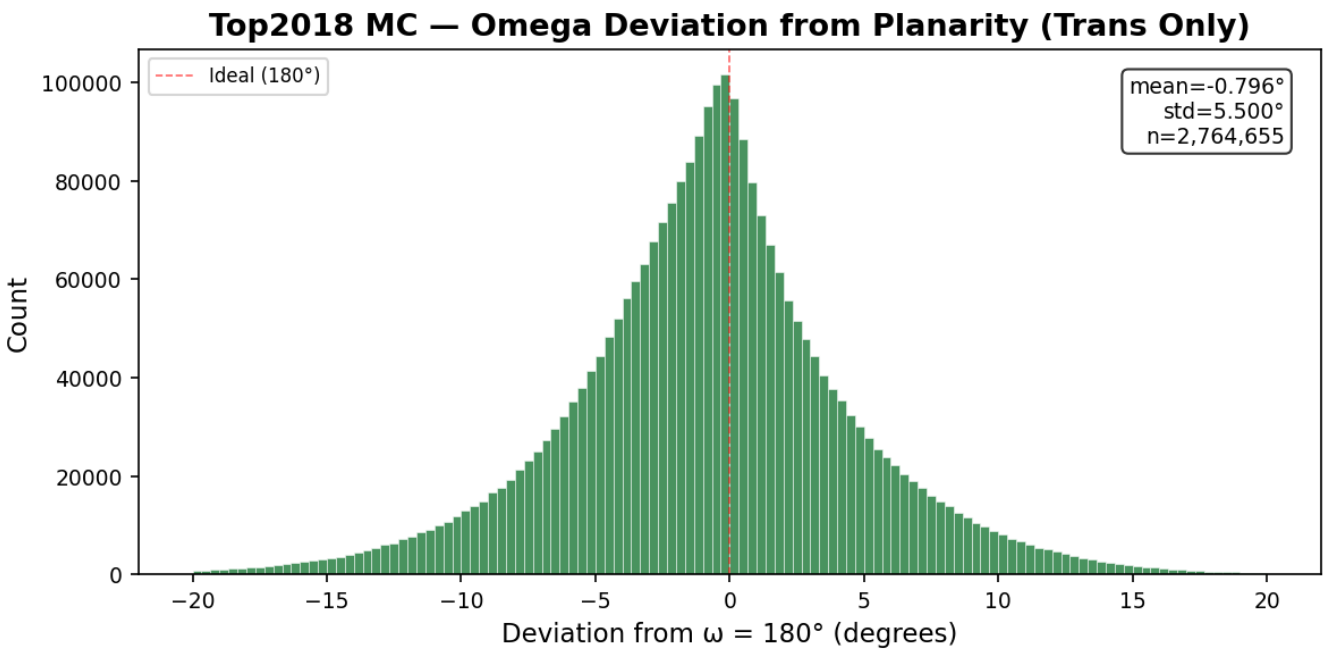
Torsion angles (φ, ψ, ω, χ) are periodic variables defined on the interval [−180°, +180°]. Standard (linear) arithmetic means and standard deviations give misleading results when values cluster near the ±180° boundary — for example, a linear mean of trans peptide bond ω values near +179° and −179° yields ~0° rather than the correct ~180°. All torsion angle summary statistics in this report use **circular statistics**: the circular mean is computed as atan2(⟨sin θ⟩, ⟨cos θ⟩), and the circular standard deviation as $\sqrt{(-2 \ln \bar{R})}$ where $\bar{R}$ is the mean resultant length.

3.1 Univariate Distributions

3.1.1 Backbone Torsion Angles (φ, ψ, ω)



3.1.2 Omega Deviation from Planarity (Trans Peptides Only)



3.1.3 Sidechain Torsion Angles (χ1–χ4)

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No chi angle data available in this dataset.
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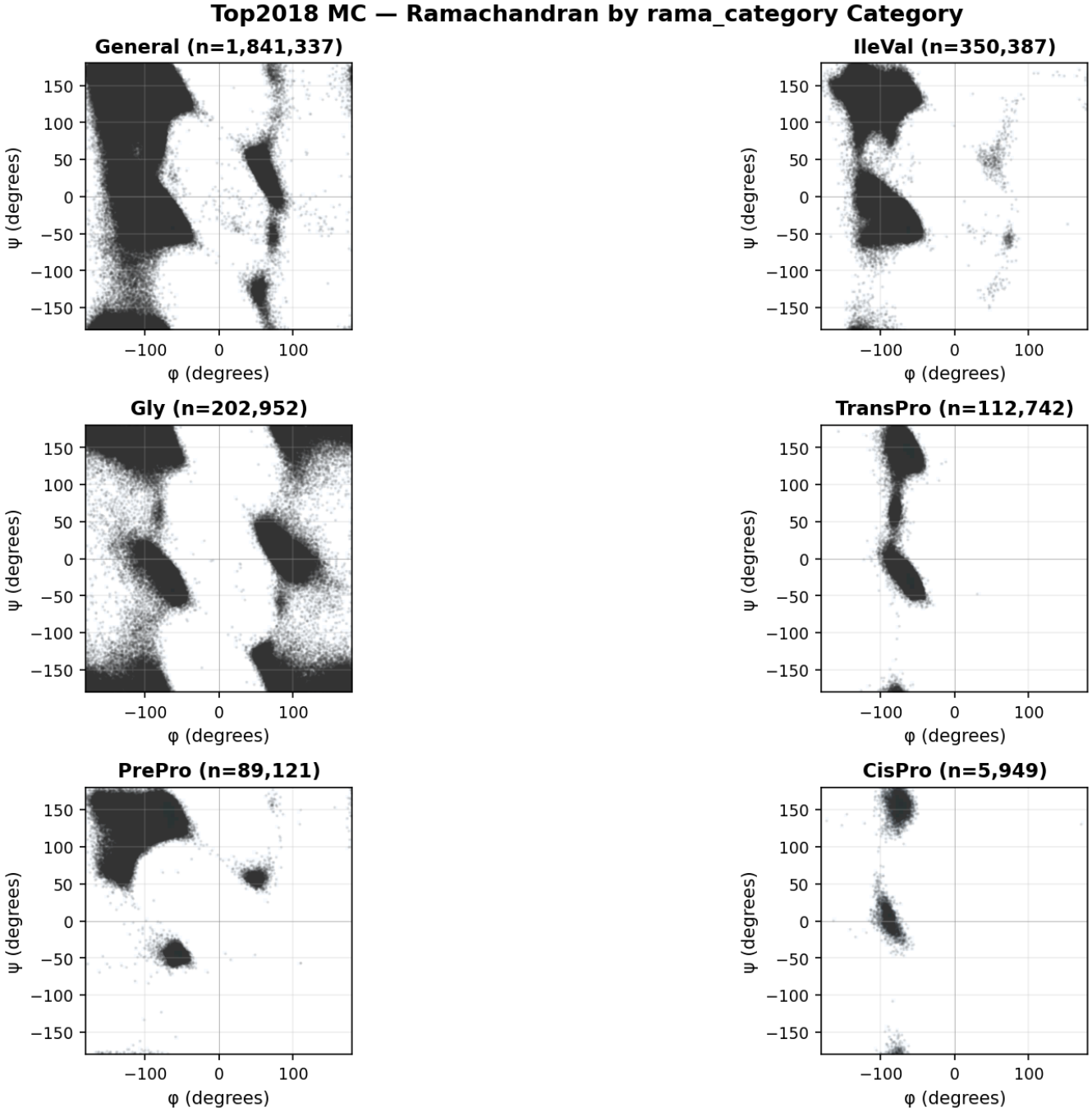
3.2 Multivariate Distributions

3.2.1 Ramachandran Distributions (φ × ψ by Category)

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Category counts (rama_category):
  General      1,841,337  (62.14%)
  IleVal         350,387  (11.82%)
  Gly            202,952  ( 6.85%)
  TransPro       112,742  ( 3.80%)
  PrePro          89,121  ( 3.01%)
  CisPro           5,949  ( 0.20%)
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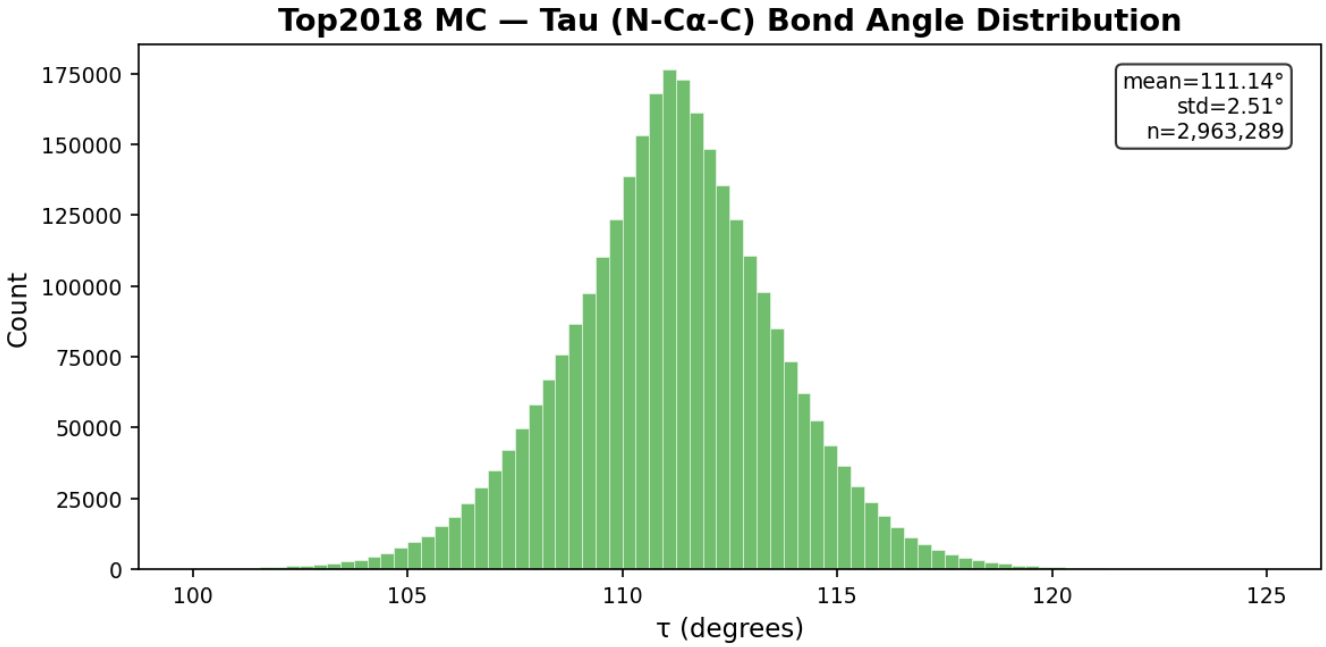
4. Bond Angle Distributions (Linear Statistics)

Bond angles are non-periodic and are analyzed with standard linear statistics (arithmetic mean, standard deviation).

4.1 Tau (N−Cα−C)



5. References

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